LINUX PROCESS MONITORING AND CONTROL SYSTEM

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Abstract

- The Linux Process Monitoring and Control System provides a unified command-line environment to inspect active Linux processes and perform basic process-control operations. It displays PID, PPID, process state and parent–child relationships using Linux interfaces such as /proc and POSIX system calls.

  The project helps learners understand process creation, execution, monitoring and control.

Problem Statement & Objectives

- Problem: Linux process details and control operations often require separate commands and interfaces, making them difficult for beginners to understand in one place.
  Objective 1: Display process details including PID, PPID and process state.
  Objective 2: Demonstrate process creation and parent–child relationships.
  Objective 3: Perform supported process-control operations using Linux interfaces and signals.
  Scope: Educational CLI-based monitoring and basic control of Linux processes.

Existing vs Proposed System

Aspect	Basic / Existing Approach	Proposed Project
Process handling	Separate commands and tools	Unified process-monitoring CLI
Communication	Process details viewed separately	Uses Linux process interfaces and system calls
Resource handling	Limited process visibility	Displays PID, PPID and state
Error handling	Manual command interpretation	Input validation and status output
Key gap	Difficult for beginners to connect concepts	One environment for learning process management

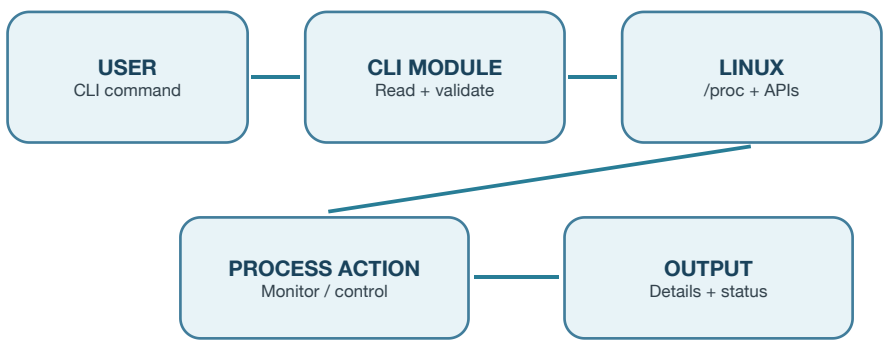
Feasibility Study

- Technical: Implemented with C/C++, GCC and Linux/Ubuntu process interfaces, including /proc and relevant system calls.
  Economic: Uses open-source tools and an existing computer; no paid hosting or software is required.
  Operational: Command-line interaction keeps the system lightweight. Users need a Linux environment and suitable permissions.
  Schedule: Developed as an academic PBL project within the semester timeline.

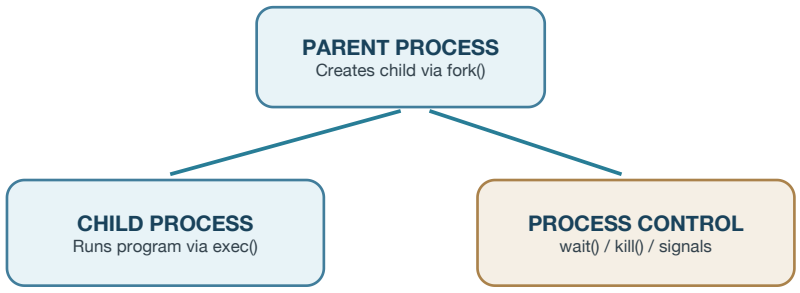
Tools & Technologies

- Programming: C / C++
  Platform: Linux / Ubuntu
  OS Concepts: Process creation, process states, parent–child relationships and process control
  System Calls / APIs: fork(), exec(), wait(), kill()
  Process Interface: Linux /proc filesystem and POSIX/Linux signals
  Tools: GCC, Terminal, VS Code, Git & GitHub

System Workflow / Architecture

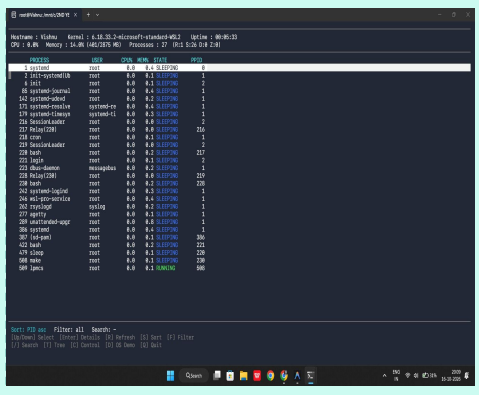


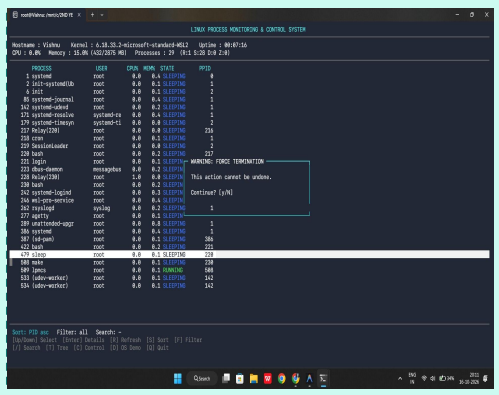
Process, IPC & Resource Design



Conceptual process relationship (only supported operations shown)

Implementation Screenshots





Results & Testing

- Designed to display PID, PPID and process state, and to show process-operation status.
  Tests: process identification, process-state retrieval, parent–child creation, supported signal/control operations and invalid-input handling.

Conclusion

- The project brings Linux process monitoring and basic control into one CLI workflow. It demonstrates process IDs, states, parent–child relationships and system-call-based process management.

Future Scope

- Real-time process dashboard and refresh.
  CPU and memory usage analytics.
  Safer priority and control options.
  Event logging, alerts and visual reports.

References: [1] Linux man-pages [2] Operating Systems course material [3] GCC / GNU documentation